

3. Square detector (1011) :-

- > Correct sequence detection:- Correct sequence should set the output high.
- > overlapping sequences:- Real machine should identify overlaps.
eg) 1011011 $\rightarrow$ 2 outputs
- > Reset conditions:- Verify reset conditions which should clear states & outputs.

> Directed tests:-

- $\rightarrow$ 1011 $\Rightarrow$ output = 1
- $\rightarrow$ 101 $\rightarrow$ reset $\rightarrow$ 1 $\Rightarrow$ output = 0
- $\rightarrow$ 1011011 $\Rightarrow$ output = 1 at cycle 1 & 7
- $\rightarrow$ 101011 $\Rightarrow$ output = 1 at cycle 6

> Coverage plan:-

- $\rightarrow$ input (0 & 1)
- $\rightarrow$ output (0 & 1)
- $\rightarrow$ lines for 0001, 0010, 0101, 1011
- $\rightarrow$ cross coverage